

Direct, Indirect, and Conditional Effects

Teaching note

Bhattacharya, Ecker, Olsson, and Schipper (2012), *The Accounting Review*
with Hayes and Rockwood (2017) and Lennox and Payne-Mann (2026)

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

Bhattacharya et al. decompose a correlation

"We use path analysis to decompose the correlation between the source (causal) variable earnings quality and the outcome variable cost of equity into direct and indirect (mediated) paths."

Bhattacharya, Ecker, Olsson, and Schipper (2012, p. 452)

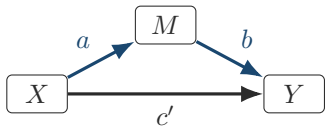
Their Table 2 (p. 463) splits the correlation of 0.2264 between accruals quality and the cost of equity into a direct path of 0.1827 (80.69 percent) and a mediated path through price impact of $0.1627 \times 0.2686 = 0.0437$ (19.31 percent).

- The note asks what this split estimates, how each piece is tested, and when mediation or moderation answers a research question.

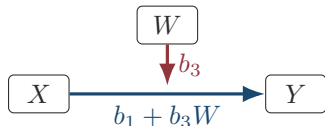
Mediation asks how; moderation asks when

"It is just as important to understand how such effects operate and the boundary conditions of those effects."

Hayes and Rockwood (2017, p. 39)



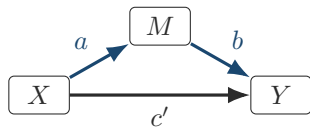
Mediation: indirect effect ab , direct effect c' .



Moderation: conditional effect
 $\theta_X(W) = b_1 + b_3W$.

- Mediation quantifies how much of the effect of X on Y travels through M ; moderation quantifies how that effect varies with W .

Mediation is estimated with three regressions



c : the coefficient of Y on X alone, eq. (1).

Hayes and Rockwood (2017, p. 41):

$$\hat{Y} = i_Y + c X \quad (1) \quad \text{the total effect of } X \text{ on } Y$$

$$\hat{M} = i_M + a X \quad (2) \quad \text{the effect of } X \text{ on } M$$

$$\hat{Y} = i_Y + c' X + b M \quad (3) \quad c' \text{ with } M \text{ held fixed; } b \text{ with } X \text{ held fixed}$$

The indirect effect ab is “the estimated difference in Y resulting from a one-unit change in X through a sequence of causal steps in which X affects M , which in turn affects Y ” (p. 43).

Covariates enter every equation; distributing them unevenly typically breaks the identity on the next slide (p. 42; fn. 2, p. 43).

- Two regressions supply a and b , whose product is the indirect effect; the third supplies the total effect c .

The total effect is the sum of two paths

Residuals e_M and e_Y from (2) and (3); intercepts collected in a constant.

$$\hat{Y} = i_Y + cX \quad (1) \quad \hat{M} = i_M + aX \quad (2) \quad \hat{Y} = i_Y + c'X + bM \quad (3)$$

Substitute $M = i_M + aX + e_M$ from (2) into (3):

$$Y = \text{const} + (c' + ab)X + (be_M + e_Y)$$

Both residuals are orthogonal to X in the same sample, so regressing Y on X returns $c = c' + ab$.

c : total effect
nothing held fixed

c' : direct effect
 M held fixed

$ab = c - c'$: indirect effect
the product of the two
segments (p. 43)

- With the same sample and covariates in every equation, $c = c' + ab$ holds exactly, so a mediated share ab/c is arithmetic, not a test.

The decomposition holds even when c is zero

```
set obs 5000
gen x = rnormal()
gen m = 0.5*x + rnormal()
gen ey = rnormal()
gen yA = 0.2*x + 0.8*m + ey
gen yB = -.4*x + 0.8*m + ey
reg m x           // a
reg yA x m        // c' and b
reg yA x          // c
```

	Design A	Design B
<i>a</i> : M on X	0.48 (34.8)	0.48 (34.8)
<i>b</i> : Y on M , X held fixed	0.79 (55.5)	0.79 (55.5)
c' : Y on X , M held fixed	0.20 (12.9)	-0.40 (-25.5)
ab (indirect)	0.38 [29.5]	0.38 [29.5]
$c' + ab$	0.59	-0.01
c : Y on X (total)	0.59 (32.8)	-0.01 (-0.8)

Coefficients (t -statistics), delta-method (Sobel) z in brackets; seed 2012, $N = 5,000$, one draw of X , e_M , and e_Y for both designs;

scripts/mediation_moderation_identity.do.

- Design B pairs an indirect effect of 0.38 with a total effect of -0.01 , “since the size of c does not determine or constrain the size of ab ” (Hayes and Rockwood 2017, p. 43).

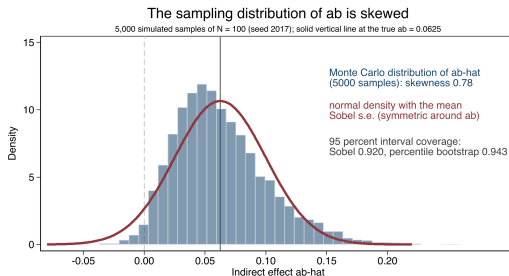
The product ab has a skewed distribution

In 5,000 samples of $N = 100$ (true $ab = 0.0625$), the estimate of ab has skewness 0.78.

Sobel interval $ab \pm 1.96 \text{ se}$: coverage 0.920; rejects zero in 24.5 percent.

Percentile bootstrap (1,000 resamples): coverage 0.943; rejects in 44.5 percent.

- A symmetric interval misplaces its tails; the percentile bootstrap follows the sampling distribution (Hayes and Rockwood 2017, p. 44).



The effect of X on Y is a line in W

Hayes and Rockwood (2017, p. 47), eq. (10):

$$\hat{Y} = i_Y + b_1X + b_2W + b_3XW$$

Differentiate with respect to X (we write $\theta_X(W)$ for the conditional effect); then write $W = W_c + \bar{W}$ with W_c the deviation from the mean:

$$\theta_X(W) = \frac{\partial \hat{Y}}{\partial X} = b_1 + b_3W \quad \hat{Y} = i_Y'' + (b_1 + b_3\bar{W})X + b_2W_c + b_3XW_c$$

b_3

does the effect of X depend
on W : the test of moderation

$b_1 = \theta_X(0)$

the effect of X at $W = 0$, not
a main effect (pp. 49–50)

centered $b_1 = \theta_X(\bar{W})$

b_3 and its t unchanged (p. 49;
Week 3, Jennings)

- The coefficient on XW tests moderation; b_1 and b_2 are conditional effects at zero, and centering W relabels b_1 as $\theta_X(\bar{W})$ with b_3 unchanged.

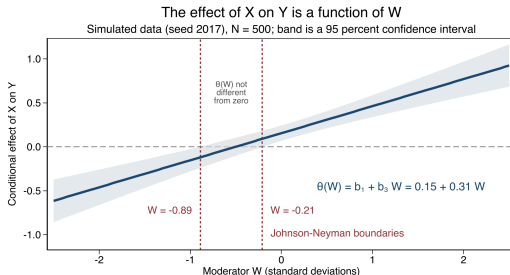
Johnson-Neyman probes the conditional effect

Pick a point: $\theta_X(-1) = -0.15$
($t = -2.4$); $\theta_X(0) = 0.15$ (3.5);
 $\theta_X(1) = 0.46$ (7.3).

Johnson-Neyman: distinguishable from zero for $W < -0.89$ and $W > -0.21$.

Median split: slopes -0.06 ($t = -0.9$) and 0.37 (5.9); this contrast is not a test of b_3 ($t = 6.7$).

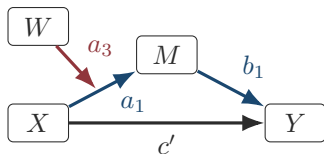
- Johnson-Neyman boundaries come from the fitted model; sample cutpoints and subgroup splits do not test b_3 (Hayes and Rockwood 2017, pp. 48, 51).



An indirect effect can itself be conditional

“As an indirect effect (mediation) is an effect, and effects can be contingent (moderation), it follows that an indirect effect can be contingent.”

Hayes and Rockwood (2017, p. 52)



Eqs. (13) and (14) (p. 53), covariates suppressed:

$$\hat{M} = i_M + a_1X + a_2W + a_3XW \quad \hat{Y} = i_Y + c'X + b_1M$$

Conditional process analysis (p. 52): X affects M by $a_1 + a_3W$ and M affects Y by b_1 , so the conditional indirect effect, eq. (17), is

$$\omega(W) = (a_1 + a_3W) b_1 = a_1b_1 + a_3b_1 W$$

$a_3b_1 = \partial\omega/\partial W$: the index of moderated mediation.

- When W moderates the a path, the indirect effect is a line in W whose slope a_3b_1 is the quantity to test.

The index of moderated mediation is the test

$N = 500$, true $\omega(W) = 0.30 + 0.18W$:

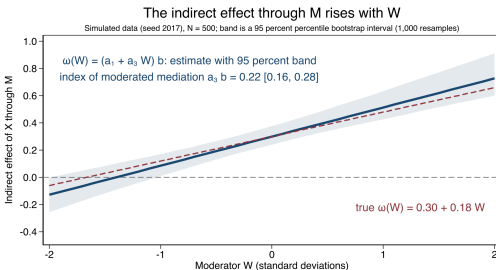
$a_1 = 0.50$, $a_3 = 0.36$ ($t = 7.8$),

$b = 0.61$ ($t = 13.8$).

$\omega(0) = a_1b = 0.31$; index $a_3b = 0.22$,
bootstrap interval $[0.16, 0.28]$.

Bhattacharya et al. test such a hypothesis by annual median splits (their Table 6, p. 475); here it is an interval on the index.

- “What matters is the index of moderated mediation and whether it is different from zero” (Hayes and Rockwood 2017, p. 53), not a_3 and b separately.



The identity fixes the sum, not the split

Lennox and Payne-Mann's (2026, p. 293) correlated-errors case, with an unobserved U carrying the correlation and X exogenous:

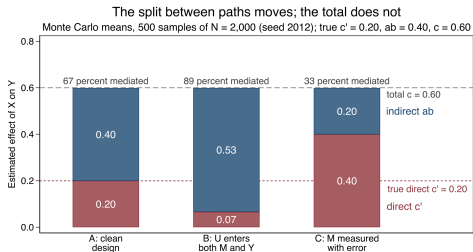
$$M = aX + \phi U + e_M,$$

$$Y = c'X + bM + \psi U + e_Y.$$

$$\hat{b} \rightarrow b + \frac{\psi\phi}{\phi^2 + \sigma_{e_M}^2}, \quad \hat{c}' \rightarrow c' - a \frac{\psi\phi}{\phi^2 + \sigma_{e_M}^2}$$

With $M^* = M + u$ instead: $\hat{b} \rightarrow \lambda b$ and $\hat{c}' \rightarrow c' + ab(1 - \lambda)$, where $\lambda = \sigma_{e_M}^2 / (\sigma_{e_M}^2 + \sigma_u^2)$; $\text{var}(U) = 1$ throughout.

- Either bias moves the total between the paths while $\hat{c}' + \hat{a}\hat{b} = c$ still holds; an exogenous X fixes the total effect, not its division.



Bhattacharya et al. report c' , ab , and c

Bhattacharya et al. Table 2 (p. 463): $0.2264 = 0.1827 + 0.1627 \times 0.2686$, with $t = 4.29$ on the product and no stated standard error. $p[EQ, Impact] = 0.1627$ in their Tables 2 to 4 (pp. 463, 465, 468): controls enter only the cost-of-equity equation, and the mediated share becomes 12.12 percent (Table 4, p. 468).

Footnote 26 concedes the second: “measurement error in a mediator variable in a path analysis can induce an underestimation of the importance of the mediated path” (p. 467). The first, no unmeasured common cause of Impact and the cost of equity, is never named.

- The tables decompose an association; reading 19 percent as a channel needs an Impact measure unconfounded with the cost of equity and error free.

The question determines the design

Question	Regressions	Estimand and report	A causal reading requires
How does X affect Y ?	mediation: M on X ; Y on X and M	ab and c' ; percentile bootstrap interval on ab	X exogenous given covariates; M unconfounded with Y , error free; X before M before Y
When, or for whom?	moderation: Y on X , W , XW	$\theta_X(W) = b_1 + b_3W$; a Johnson-Neyman plot	X exogenous given covariates; the linear-in- W form; W well measured
How, depending on when?	conditional process: M on X , W , XW ; Y on X , W , XW , M	$\omega(W) = (a_1 + a_3W)b_1$; interval on the index a_3b_1 ; Bhattacharya et al. Table 6 (p. 475) splits at the median instead	both of the above
Is a causal split available?	the same, when neither uncorrelated errors nor an exclusion restriction is defensible	c' and ab as OLS associations, not effects	Lennox and Payne-Mann (2026, p. 303): "refrain from using path analysis to obtain evidence of causal processes"

- Mediation and moderation answer different questions; the regression supplies neither the ordering nor the absence of confounding that a causal reading needs.

Summary

- Mediation and moderation answer different questions: the indirect effect ab quantifies how much of the effect of X on Y travels through M ; the conditional effect $b_1 + b_3W$ quantifies how that effect varies with W ; when the mechanism depends on W , the index a_3b_1 is the quantity to test.
- Under OLS on one sample the total effect decomposes exactly as $c = c' + ab$, so a mediated share is arithmetic rather than a test and a null total effect does not preclude an indirect effect; the product ab has a skewed sampling distribution that favors a percentile bootstrap interval over a normal-theory test.
- The causal reading of any split, in Bhattacharya et al. (2012) as elsewhere, rests on the mediator being unconfounded with the outcome and measured without error; an exogenous treatment fixes the total effect but not its division between paths.

Bhattacharya, Ecker, Olsson, and Schipper (2012) · Hayes and Rockwood (2017) · Lennox and Payne-Mann (2026) · github.com/sbb151/acctg597a-research-design